

Microeconomics

Lecture 1: Thinking Like an Economist

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Course roadmap

Course structure

1. Thinking like economists
2. Consumer theory
3. Production, technology and costs
4. Competitive markets and welfare
5. Market power (monopoly)
6. Oligopoly and strategic interaction
7. Asymmetric information
8. Market failures and public policy

Today's objectives

1. **Understand** what it means to “think like an economist”
2. **Recognize** economic principles in everyday choices
3. **Distinguish** between:
 - ▶ sunk costs vs opportunity costs
 - ▶ marginal analysis vs total analysis
 - ▶ micro vs macroeconomics
4. **Essential math review** (tools for the whole course)
5. **Link** to consumer theory

An opening question

Why does the price of a Milan–Palermo flight vary so much?

Same flight, same airline, but:

- booking 3 months ahead: €50
- booking 1 week ahead: €250
- booking the day before: €500

Economic answer (idea):

- those who book early are often **more price-sensitive**
- those who book at the last minute are often **less price-sensitive**
- the airline practices **price discrimination** to increase profits

Thinking like economists: looking for the logic of incentives behind seemingly “random” phenomena.

Microeconomics: a definition

Definition

Microeconomics studies the behavior of individual economic agents (consumers, firms, workers) and their interactions in markets.

Key questions:

- How do people decide what to buy given a **budget**?
- How does a firm decide how much to produce and at what price to sell?
- How is the **price** determined in a market?
- How are **scarce resources** allocated among alternative uses?

Examples of markets

- rents and housing
- labor and skills
- agricultural goods (wheat, coffee, olive oil)

Microeconomics vs Macroeconomics

Microeconomics

- individual economic agents
- specific markets
- individual decisions
- relative prices
- resource allocation

Examples:

- price of gasoline
- choice: study vs work
- a firm's pricing strategy

Macroeconomics

- the aggregate economic system
- GDP, inflation, unemployment
- economic policy
- general price level
- economic growth

Examples:

- inflation rate
- unemployment rate
- monetary policy (ECB)

Note

The two disciplines are **complementary**: macro is often built on micro foundations (choices, incentives, constraints).

Scarcity: everyone's problem

The central economic problem

Resources are scarce (limited), while **wants are potentially unlimited**.

Unavoidable consequences:

1. we must make **choices**
2. every choice involves **giving something up** (trade-offs)
3. we must **compare** costs and benefits

Everyday examples of scarcity

- **Time**: study vs work vs rest
- **Money**: rent vs food vs leisure
- **Energy**: sport vs study, etc.

Trade-offs everywhere

People (consumption and time):

- Budget €500/month: more food $\Rightarrow$ less clothing
- Saturday night: cinema (monetary cost + time) vs studying (cost of giving up relaxation)

Work:

- job A: €1,800/month, flexible hours, stimulating environment
- job B: €2,200/month, rigid hours, routine
- trade-off: wage vs quality of life

Firms and governments:

- R&D: investing in project X $\Rightarrow$ fewer resources for Y
- public budget: more health care $\Rightarrow$ less education (for given revenues)

Activity: your trade-offs (Think–Pair–Share)

3+3+2 minutes

Think (3'): about a recent important choice (study/work, a purchase, time management).

Pair (3'): compare with the person next to you.

Share (2'): 2–3 examples with the class.

Guiding questions

1. What were the realistic alternatives?
2. What did you give up?
3. What was the choice criterion (even if implicit)?

1. Opportunity cost

Definition

The **opportunity cost** of a choice is the value of the **best alternative** forgone.

It is not just the price paid: it includes time, alternative income, lost benefits.

Example: University

- **Explicit costs:** tuition (€2,000/year), books (€500), etc.
- **Opportunity cost:** wage forgone by not working (e.g. €18,000/year)
- **Economic cost** = explicit costs + value of the best alternative

Common pitfall

Confusing “free” with “costless”. If you spend time, you are paying.

Opportunity cost: quick exercise

Situation: You won a free ticket to a concert (value €50). The concert is tonight. You also have a part-time shift that pays €60.

Questions:

1. What is the opportunity cost of going to the concert?
2. What would you do? (it depends on how much you value the concert)

Answer (logic):

- The free ticket does *not* eliminate the cost: the opportunity cost is €60 (forgone wage).
- Go to the concert only if the perceived benefit is $> €60$.

2. Sunk costs

Definition

Sunk costs are costs already incurred that **cannot be recovered**, whatever decision is made now.

Golden rule

Sunk costs **should not** affect future decisions: they are in the past.

Example: a boring movie

- You paid €10. After 30 minutes the movie is terrible.
- Is staying “because I paid” rational?

Economic answer: NO. The decision depends on **costs and benefits**.

Sunk cost fallacy: the trap

Continuing an activity only because time/money has already been invested, even if it is **no longer worthwhile**.

Examples:

1. Unused **gym membership**: “I have to go so it doesn’t go to waste”.
2. A **project** that isn’t working: “I’ve invested too much by now”.
3. **Consumption choice**: “I’ve already bought it, so I have to use it”.

Key message

Optimal choices look to the future: **future benefits and costs**.

Exercises: spotting a sunk cost

Scenario 1: You bought a book (€60). You find the same content for free in a better resource. What do you do?

Answer: the €60 is sunk. Use the better resource (today).

Scenario 2: Monthly transit pass already paid. Today it's raining and a taxi would save you time. What do you do?

Answer: the pass is sunk. Compare today: taxi cost vs value of the time saved.

Scenario 3: You studied 20 hours on the wrong syllabus. You realize it the day before the exam. What do you do?

Answer: the 20 hours are sunk. Now it pays to reallocate study to the correct syllabus.

3. Marginal analysis

Principle

Rational decisions are based on comparing **marginal benefit** (MB) and **marginal cost** (MC).

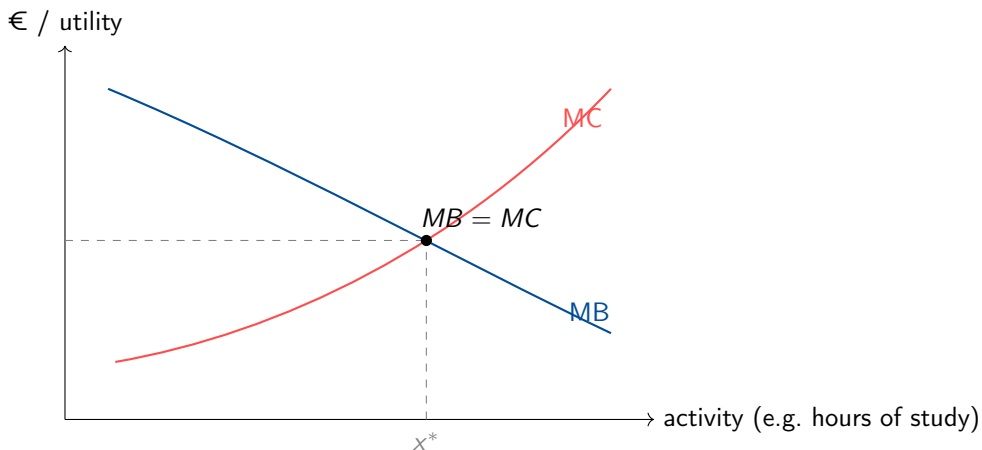
Rule:

- increase an activity if $MB > MC$
- reduce/stop if $MB < MC$
- optimum when $MB = MC$

Example: hours of study

Hour	MB (points)	MC (effort)	Worth it?
1	10	3	Yes
2	8	3	Yes
3	5	3	Yes
4	3	3	Indifferent
5	1	3	No

Marginal analysis: graph (MB and MC)



Intuition

The point x^* is “optimal” because before x^* marginal gains exceed marginal costs; after it, the opposite holds.

Example: a firm and production

Example: producing pizzas (price €8)

Pizza no.	Revenue	Total cost	MC	Marg. profit
1	8	3	3	+5
2	8	6	3	+5
3	8	10	4	+4
4	8	15	5	+3
5	8	21	6	+2
6	8	29	8	0
7	8	39	10	-2

Conclusion: produce as long as marginal profit is non-negative (here: 6 pizzas).

Summary of Part 1

What we did:

- Microeconomics: individual choices and markets
- Scarcity $\Rightarrow$ trade-offs
- Three key ideas:
 - ▶ opportunity cost
 - ▶ sunk costs
 - ▶ marginal analysis

— 10-MINUTE BREAK —

Next step

Now we need the **mathematical tools** to formalize and solve economic problems.

Why mathematics in economics?

Mathematics is the language of modern economics.

Why we need it:

- **precision:** it removes ambiguity
- **rigor:** it allows verifiable arguments
- **modeling:** it represents complex relationships
- **solving:** it lets us compute optimal choices and equilibria

Important

You don't need to be "good" at math: you need to understand the **logic** of the tools and practice with exercises.

Functions: basic concept

Definition

A **function** $y = f(x)$ assigns to each value of x (input) a unique value of y (output).

Interpretation:

- x : independent variable (explanation/cause)
- y : dependent variable (outcome/effect)
- f : relationship between x and y

Economic examples:

- demand: $Q_d = 100 - 2P$
- utility: $U(x, y) = \sqrt{xy}$
- total cost: $C(Q) = 100 + 5Q$

Linear functions

Form: $y = a + bx$

Parameters:

- a = intercept (value of y when $x = 0$)
- b = slope: how much y changes if x increases by 1

Example: $Q_d = 100 - 2P$

- intercept: 100
- slope: -2 (if $P \uparrow 1$, then $Q_d \downarrow 2$)

Slope: computation and interpretation

Formula

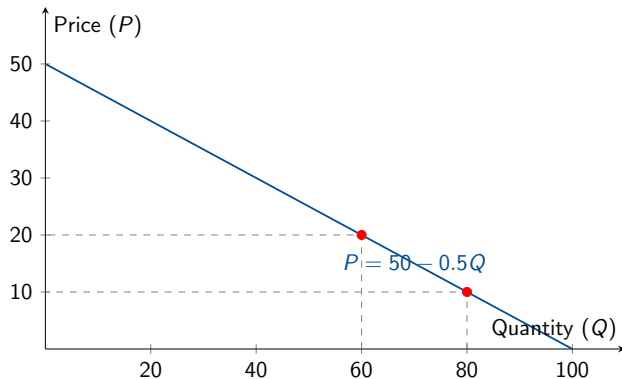
$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Example: points $(P_1, Q_1) = (10, 80)$ and $(P_2, Q_2) = (20, 60)$

$$\text{Slope} = \frac{60 - 80}{20 - 10} = -2$$

Interpretation: “If the price increases by 1, the quantity decreases by 2”.

Graph: demand curve



Derivatives: intuition

What does the derivative measure?

The derivative $f'(x)$ measures the **instantaneous rate of change** of $f(x)$ with respect to x .

Geometric interpretation: slope of the tangent line.

Economic interpretation:

- marginal utility: $\frac{dU}{dx}$
- marginal cost: $\frac{dC}{dQ}$
- marginal product: $\frac{dY}{dL}$

Fundamental rules of differentiation

Basic rules

1. constant: $f(x) = c \Rightarrow f'(x) = 0$
2. power: $f(x) = x^n \Rightarrow f'(x) = nx^{n-1}$
3. logarithm: $f(x) = \log(x) \Rightarrow f'(x) = \frac{1}{x}$
4. sum/difference: $(f \pm g)' = f' \pm g'$
5. constant times a function: $(cf)' = cf'$

Examples:

$$7 \Rightarrow 0$$

$$x^4 \Rightarrow 4x^3$$

$$3x^2 \Rightarrow 6x$$

$$x^2 + 5x - 3 \Rightarrow 2x + 5$$

$$x^{-1} \Rightarrow -x^{-2}$$

$$x^{1/2} \Rightarrow \frac{1}{2}x^{-1/2}$$

Product rule

Rule

If $f(x) = g(x)h(x)$, then

$$f'(x) = g'(x)h(x) + g(x)h'(x)$$

Example: $f(x) = x^2(3x + 5)$

$$f'(x) = 2x(3x + 5) + x^2 \cdot 3 = 9x^2 + 10x$$

Quotient rule

Rule

If $f(x) = \frac{g(x)}{h(x)}$, then

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{h(x)^2}$$

Example: $f(x) = \frac{x^2+1}{x}$

$$f'(x) = \frac{2x \cdot x - (x^2 + 1) \cdot 1}{x^2} = 1 - \frac{1}{x^2}$$

Chain rule

Rule

If $f(x) = g(h(x))$, then

$$f'(x) = g'(h(x)) \cdot h'(x)$$

Example: $f(x) = (2x + 3)^5$

$$f'(x) = 5(2x + 3)^4 \cdot 2 = 10(2x + 3)^4$$

Maxima and minima: conditions

First-order condition (FOC)

A candidate maximum/minimum x^* satisfies:

$$f'(x^*) = 0$$

Second-order condition (SOC)

- if $f''(x^*) < 0$, then x^* is a maximum
- if $f''(x^*) > 0$, then x^* is a minimum

Economic example: maximizing profit

Profit function: $\pi(Q) = 120Q - 2Q^2 - 200$

FOC:

$$\pi'(Q) = 120 - 4Q = 0 \Rightarrow Q^* = 30$$

SOC:

$$\pi''(Q) = -4 < 0 \Rightarrow \text{maximum}$$

Maximum profit:

$$\pi(30) = 1,600$$

Guided exercise: derivatives

1. $f(x) = 7x^4 - 3x^2 + 5x - 10$

Solution: $f'(x) = 28x^3 - 6x + 5$

2. $g(x) = (x^2 + 1)(3x - 2)$

Solution: $g'(x) = 9x^2 - 4x + 3$

3. $h(x) = \frac{x^2 - 4}{x + 2}$

Solution: $h(x) = x - 2$ (for $x \neq -2$) $\Rightarrow h'(x) = 1$

4. $k(x) = (3x^2 - 1)^4$

Solution: $k'(x) = 24x(3x^2 - 1)^3$

Linear equations

Form: $ax + b = 0 \Rightarrow x = -\frac{b}{a}$ (if $a \neq 0$)

Application: market equilibrium

$$Q_d = 100 - 2P, \quad Q_s = 20 + 3P$$

In equilibrium $Q_d = Q_s$:

$$100 - 2P = 20 + 3P \implies P^* = 16, \quad Q^* = 68$$

Quadratic equations

Form: $ax^2 + bx + c = 0$

Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example: $x^2 - 7x + 10 = 0 \Rightarrow x = 5$ or $x = 2$.

2×2 systems

$$\begin{cases} 2x + y = 12 \\ x - y = 3 \end{cases}$$

Substitute y from the second equation ($y = x - 3$) into the first:

$$2x + (x - 3) = 12 \Rightarrow x = 5, \quad y = 2$$

Math summary

Tools acquired:

- linear functions (slope/intercepts)
- derivatives (basic rules, product, quotient, chain)
- maxima/minima (FOC+SOC)
- equations and systems

— 5-MINUTE BREAK —

Next step

Let's apply what we have seen through some exercises

Case 1: university or work?

Situation: a person must decide between university (3 years) and working right away.

University:

- explicit costs: €2,500/year + €500/year books $\Rightarrow$ €9,000
- opportunity cost: €18,000/year for 3 years $\Rightarrow$ €54,000
- total economic cost: €63,000

Benefit: future salary €35,000/year vs €18,000 without a degree.

In how many years does the higher annual income recover the cost incurred?

$$\frac{63,000}{35,000 - 18,000} \approx 3.7 \text{ years}$$

Case 2: produce or shut down? (sunk costs)

Situation: fixed costs already paid €10,000. Producing 100 units:

- variable cost: €80/unit $\Rightarrow$ €8,000
- price: €90/unit $\Rightarrow$ revenue €9,000

Correct analysis:

- the €10,000 is **sunk** (you pay it anyway)
- compare **only** revenue vs variable cost: €9,000 vs €8,000
- producing pays: you reduce the total loss (from €10,000 to €9,000)

Case 3: optimal allocation of study time

Situation: You have 5 hours in total to split between Exam A and Exam B.

Objective: Maximize the total score:

$$\text{Total} = P_A(h_A) + P_B(h_B)$$

subject to

$$h_A + h_B = 5$$

Marginal benefits (additional points for each extra hour):

Hours studied	MB A	MB B
1	5.0	4.0
2	4.5	4.0
3	4.2	4.0
4	3.8	4.0
5	3.8	3.5

Note: Marginal benefits are decreasing. The second hour is worth less than the first, etc.

Case 3: applying the marginal rule

We compare marginal benefits hour by hour.

Hour 1: A ($5.0 > 4.0$)

Hour 2: A ($4.5 > 4.0$)

Hour 3: A ($4.2 > 4.0$)

Hour 4: B ($4.0 > 3.8$)

Hour 5: B ($4.0 > 3.8$)

Optimal allocation:

$$h_A = 3 \quad h_B = 2$$

Total score:

$$(5 + 4.5 + 4.2) + (4 + 4) = 21.7$$

General rule:

Optimal allocation when $MB_A \approx MB_B$

Case 4: estimating a linear demand

Data:

Price (€)	Quantity (cups/day)
1.00	120
1.50	100
2.00	80
2.50	60
3.00	40

Estimate (linear): $Q_d = a - bP$

- $-b = \frac{\Delta Q}{\Delta P} = \frac{100-120}{1.5-1.0} = -40 \Rightarrow b = 40$
- using (1, 120): $120 = a - 40 \Rightarrow a = 160$

$$Q_d = 160 - 40P$$

Full exercise: equilibrium and a tax

Market for good X:

$$Q_d = 200 - 10P, \quad Q_s = -50 + 15P$$

1) Initial equilibrium

$$200 - 10P = -50 + 15P \implies P^* = 10, \quad Q^* = 100$$

2) Tax of €2/unit on producers

$$Q_s = -50 + 15(P - 2) = -80 + 15P$$

$$200 - 10P = -80 + 15P \Rightarrow P = 11.2, \quad Q = 88$$

Preview: the consumer's optimal choice

Utility: $U(x, y) = xy$

Constraint: $2x + 4y = 120$

Idea: we turn the constraint into a function and maximize.

$$y = 30 - 0.5x \Rightarrow U(x) = x(30 - 0.5x) = 30x - 0.5x^2$$

$$U'(x) = 30 - x = 0 \Rightarrow x^* = 30, \quad y^* = 15$$

The “recipe” of Consumer Theory

Three ingredients (always)

1. **Preferences:** what one likes (indifference curves)
2. **Constraint:** what is feasible (budget line)
3. **Optimum:** where one chooses (tangency / marginal conditions)

Translation into economic language

Optimal choice when the subjective trade-off (**MRS**) and the market trade-off (relative prices) coincide.

Take-home points from today

- ✓ Economists reason in terms of **trade-offs** and **incentives**.
- ✓ The relevant cost is often the **opportunity cost**, not the accounting cost.
- ✓ **Sunk costs** should not guide future choices.
- ✓ Optimal decisions are understood through **marginal analysis**.
- ✓ Mathematics makes this reasoning **precise and solvable**.

Self-check questions

1. What is the difference between opportunity cost and sunk cost?
2. Why should sunk costs not affect future decisions?
3. How do you find a maximum using the derivative? (FOC and SOC)
4. What does “marginal analysis” mean?
5. Given $Q_d = 80 - 2P$ and $Q_s = 20 + 3P$, find (P^*, Q^*) .